

Differential Equations for Master Integrals

Lecture notes on the analytic evaluation of the double-real phase-space integrals in
 $pp \rightarrow h + X$ at NNLO

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Abstract

Integration-by-parts reduction expresses all phase-space integrals of the double-real channel through a finite basis of master integrals. These notes develop, from first principles, the method by which the masters are evaluated: the derivation of first-order differential equations in the kinematic invariants; the organization of the equations into triangular systems; the identification of integral families by changes of loop momenta and, separately, of the diagonal blocks of the resulting systems by conjugation, which together reduce the many systems to a short list of genuinely distinct ones; the canonical form of the equations, in which the Laurent expansion in the dimensional regulator integrates order by order into iterated integrals; the treatment of square roots of kinematic polynomials by rational changes of variables and, where no such change of variables exists, by the Galois theory of the roots; the determination of boundary conditions from the analytic and geometric properties of the phase space; and the recovery of the distributional behaviour of the integrals at the edges of phase space.

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1 Introduction

1.1 Phase-space integrals as cut loop integrals

The physical quantity behind these notes is the NNLO hard function for single-inclusive hadron production, $pp \rightarrow h + X$, in collinear factorization. In its double-real channel two unresolved massless partons are radiated and integrated over, so the basic objects are phase-space integrals of squared tree amplitudes. It is convenient to convert them into loop integrals at the outset. By the Sokhotski–Plemelj identity,

$$\delta^{(+)}(p^2) = \frac{1}{2\pi i} \left[\frac{1}{p^2 - i0} - \frac{1}{p^2 + i0} \right] \quad \text{on positive-energy configurations,} \quad (1)$$

each on-shell delta function is the discontinuity of a propagator; the positive-energy condition $\theta(p^0)$ is supplied by the cut itself rather than by the discontinuity. Every phase-space integral thereby becomes a two-loop integral in which certain lines are *cut*. Cut integrals obey the same integration-by-parts identities as ordinary loop integrals, because the delta functions are ordinary distributions: differentiation under the integral sign and the vanishing of total derivatives are

unimpaired. This rewriting is the device known as reverse unitarity [4]. Two rules govern a cut line. Raising its power to $a_i \geq 1$ produces

$$\frac{1}{D_i^{a_i}} \longleftrightarrow \frac{(-1)^{a_i-1}}{(a_i-1)!} \delta^{(a_i-1)}(D_i), \quad (2)$$

again an admissible integrand; lowering it to $a_i \leq 0$ removes the delta function altogether, and such an integral vanishes, having no discontinuity in that channel. The methods developed for loop integrals thereby become available for phase-space integration.

The partonic process behind the channel is

$$p_1 + p_2 \longrightarrow p_3 + k_1 + k_2, \quad (3)$$

with p_1, p_2 the incoming partons, p_3 the observed one, and k_1, k_2 the two radiated partons that are integrated over, all massless. Its invariants are

$$s = (p_1 + p_2)^2, \quad t = (p_1 - p_3)^2, \quad u = (p_2 - p_3)^2, \quad (4)$$

and the integrals depend on them only through the two dimensionless ratios

$$v = -\frac{t}{s}, \quad w = -\frac{u}{s}. \quad (5)$$

The physical region follows from the kinematics alone. Since p_3 is massless,

$$s + t + u = (p_1 + p_2 - p_3)^2 = M^2, \quad (6)$$

the squared invariant mass of the unobserved pair, so that $1 - v - w = M^2/s \geq 0$; and $v = 2p_1 \cdot p_3/s > 0$, $w = 2p_2 \cdot p_3/s > 0$. The region is therefore the open triangle

$$0 < v, \quad 0 < w, \quad v + w < 1. \quad (7)$$

In the partonic centre-of-mass frame, with E_3 the energy of the observed parton and θ the angle between $\vec{p}_3$ and $\vec{p}_1$,

$$v = \frac{E_3}{\sqrt{s}} (1 - \cos \theta), \quad w = \frac{E_3}{\sqrt{s}} (1 + \cos \theta), \quad (8)$$

so $v + w = 2E_3/\sqrt{s}$ fixes the energy and v/w the angle. The three edges of the triangle are the three degenerate limits: $v \rightarrow 0$ and $w \rightarrow 0$, in which p_3 becomes collinear to p_1 and to p_2 ; and $v + w \rightarrow 1$, the Born configuration $E_3 = \sqrt{s}/2$, $M^2 = 0$, in which the unobserved pair carries no invariant mass and the radiation is unresolved — the *soft edge*, as it will be called below. All three edges reappear throughout: as singular loci of the differential equations, as the limits at which boundary conditions are read off, and as the endpoints at which the masters turn into distributions. Everything is dimensionally regularized, $d = 4 - 2\varepsilon$.

1.2 Statement of the problem

Reduction by integration-by-parts identities leaves a basis of 347 *master integrals*. A master is not itself a function of (v, w) alone: an L -loop integral whose propagator powers sum to a has mass dimension $Ld - 2a$, and Lorentz invariance then fixes its dependence on the overall scale,

$$I_m = (-s)^{Ld/2 - a_m} \hat{I}_m(v, w; \varepsilon), \quad (9)$$

with $\hat{I}_m$ dimensionless. The differential equations below are equations for the $\hat{I}_m$, and the notation $I_m(v, w; \varepsilon)$ used from here on refers to them; the bubble of Example 2.1 is the case $L = 1$, $a = 2$, where $(-s)^{d/2-2} = (-s)^{-\varepsilon}$ and $\hat{B} = c(\varepsilon)$ is a constant. The reduction is exact linear algebra over the field $\mathbb{Q}(v, w, \varepsilon)$, but it says nothing about the values of the basis elements. The task is their *analytic evaluation*: for each master, a Laurent expansion

$$I(v, w; \varepsilon) = \sum_{k \geq k_0} \varepsilon^k I^{(k)}(v, w) \quad (10)$$

whose coefficients are explicit transcendental functions — in the end, $\mathbb{Q}$ -linear combinations of iterated integrals of logarithmic differential forms — together with exactly known constants.

The method rests on one mathematical fact: *the masters satisfy first-order linear differential equations in v and w* . The sections below derive those equations, reduce the many systems to a short list of distinct ones, construct the canonical form, treat the square roots the kinematics forces on some of them, and determine the boundary data and the endpoint behaviour.

2 Differential equations for master integrals

2.1 Masters and integration by parts

Fix a set of inverse propagators $D_1, \dots, D_N$ — including the cut ones — large enough that every scalar product of loop and external momenta is a linear combination of the D_i and external invariants. Such a set, an *integral family*, defines the integrals

$$I_{a_1 \dots a_N}(v, w; \varepsilon) = \int \prod_{\ell=1}^2 \frac{d^d k_\ell}{i\pi^{d/2}} \frac{1}{D_1^{a_1} \dots D_N^{a_N}}, \quad a_i \in \mathbb{Z}, \quad (11)$$

where for a cut line the factor $D_i^{-a_i}$ stands for the delta function of (2), and an integral with $a_i \leq 0$ on a cut line vanishes. The *sector* of an integral is the subset of propagators present with positive power; deleting propagators leads to *subsectors*, and the sectors of a family form a lattice under this deletion. Integration-by-parts identities,

$$0 = \int \prod_{\ell} d^d k_\ell \frac{\partial}{\partial k_j^\mu} \left[\frac{q^\mu}{D_1^{a_1} \dots D_N^{a_N}} \right], \quad q \in \{k_1, k_2, p_i\}, \quad (12)$$

relate integrals with shifted indices; solving them expresses every integral of the family through a finite basis of master integrals, distributed over the sectors of the family.

2.2 Derivation of the differential system

A master depends on the external momenta only through their invariants $s_{kl} = (p_k + p_l)^2$, so the chain rule gives, for each pair (i, j) of external momenta,

$$p_j^\mu \frac{\partial}{\partial p_i^\mu} I = \sum_{k \leq l} \left(p_j^\mu \frac{\partial s_{kl}}{\partial p_i^\mu} \right) \frac{\partial I}{\partial s_{kl}}, \quad (13)$$

a linear system for the derivatives $\partial I / \partial s_{kl}$ whose coefficients are again invariants. Its matrix is invertible wherever the Gram determinant of the independent external momenta is non-zero, and inverting it expresses each $\partial / \partial s_{kl}$ — hence ∂_v and ∂_w — as an explicit combination of the

operators $p_j^\mu \partial / \partial p_i^\mu$. Those may be taken under the integral sign, where they raise propagator powers and generate numerators $p_j \cdot k_\ell$ and $p_i \cdot p_j$; by the completeness assumed of the family these numerators are again linear combinations of the D_i and of invariants. Every resulting term is thus an integral of *the same family*, and integration by parts maps it back to the masters. Collecting the masters of a family and its subsectors into a vector $\vec{I}$, one obtains a closed linear system

$$\partial_v \vec{I} = A_v(v, w; \varepsilon) \vec{I}, \quad \partial_w \vec{I} = A_w(v, w; \varepsilon) \vec{I}, \quad (14)$$

with matrices of rational functions. It is convenient to combine the pair into the matrix-valued one-form $A = A_v dv + A_w dw$, the *connection*, and to write

$$d\vec{I} = A\vec{I}. \quad (15)$$

A total differential equation of this shape — one first-order equation for the differential of an unknown vector, in several variables at once — is what the literature calls a *Pfaffian system*, and (15) will be referred to as the Pfaffian system of the family. A Pfaffian system is not solvable for arbitrary A : the two equations of (14) constrain each other, and the condition under which they are compatible is derived in Section 2.4. The differential-equation method for master integrals goes back to Kotikov [1] and to Gehrmann and Remiddi [2].

Example 2.1 (The one-mass bubble). *Let*

$$B(s; \varepsilon) = \int \frac{d^d k}{i\pi^{d/2}} \frac{1}{k^2 (k+p)^2}, \quad s = p^2, \quad (16)$$

a one-scale family with $D_1 = k^2$, $D_2 = (k+p)^2$ and a single nontrivial master, B itself. Here (13) has a single term, $p^\mu \partial_{p^\mu} = 2s \partial_s$, so that $\partial_s = \frac{1}{2s} p^\mu \partial_{p^\mu}$ and

$$\frac{\partial B}{\partial s} = -\frac{1}{2s} \int \frac{d^d k}{i\pi^{d/2}} \frac{2p \cdot (k+p)}{k^2 [(k+p)^2]^2}. \quad (17)$$

The numerator is a combination of inverse propagators, $2p \cdot (k+p) = D_2 - D_1 + s$, so the right-hand side involves only integrals of the family: $I_{1,1} = B$, the scaleless $I_{0,2} = 0$, and $I_{1,2}$. The integration-by-parts identity generated by $\partial_{k^\mu}(k^\mu/D_1 D_2)$ reads $0 = (d-3) I_{1,1} + s I_{1,2}$, that is,

$$I_{1,2} = -\frac{d-3}{s} I_{1,1}, \quad (18)$$

and therefore

$$\frac{\partial B}{\partial s} = -\frac{1}{2s} \left[B - (d-3) B \right] = -\frac{4-d}{2s} B = -\frac{\varepsilon}{s} B. \quad (19)$$

The equation determines the entire s -dependence,

$$B(s; \varepsilon) = c(\varepsilon) (-s)^{-\varepsilon}, \quad (20)$$

with the branch fixed by the causal prescription $s \rightarrow s + i0$ — consistent with dimensional analysis, since B carries mass dimension $d-4 = -2\varepsilon$. A single direct evaluation fixes the remaining constant,

$$c(\varepsilon) = \frac{\Gamma(1+\varepsilon) \Gamma(1-\varepsilon)^2}{\varepsilon (1-2\varepsilon) \Gamma(1-2\varepsilon)}. \quad (21)$$

The pattern is the one followed throughout: the differential equation fixes the functional dependence, and a single elementary evaluation fixes the constant left free by it. The value of $c(\varepsilon)$ is recovered from the parametric representation in Section 3.

2.3 Triangular structure

Order the sectors of a family by their number of propagators. Differentiating an integral of a sector S raises the powers of the lines already present; the numerators so produced are rewritten through the D_i , which may cancel a line and land in a subsector of S , as it does for $I_{0,2}$ in Example 2.1. The integration-by-parts identities likewise express any integral of S through integrals of S and of its subsectors, since an identity may cancel a line but never create one. Neither operation can produce a line that was absent, so both lead sideways and down the sector lattice, never up it. With the master vector ordered accordingly, the connection is *block lower triangular*,

$$A = \begin{pmatrix} a_{11} & & & \\ a_{21} & a_{22} & & \\ \vdots & & \ddots & \\ a_{b1} & a_{b2} & \cdots & a_{bb} \end{pmatrix}, \quad (22)$$

where each diagonal entry a_{ss} — a *block*, in the terminology fixed in the next section — couples the masters of one group of sectors among themselves, and the entries below the diagonal express how lower sectors source higher ones. The diagonal blocks carry the irreducible analytic content of the system; the entries below the diagonal can be dealt with afterwards — the observation on which Section 5 rests.

2.4 Integrability

Equality of mixed partial derivatives, $\partial_v \partial_w \vec{I} = \partial_w \partial_v \vec{I}$, forces the compatibility condition

$$\partial_v A_w - \partial_w A_v - [A_v, A_w] = 0, \quad \text{equivalently} \quad dA = A \wedge A, \quad (23)$$

the statement that the connection is *flat*. This is the integrability condition of the Pfaffian system (15): it is what makes the two equations of (14) compatible, and a Pfaffian system satisfying it is called integrable. (To check the sign: $\partial_v \partial_w \vec{I} = \partial_v (A_w \vec{I}) = (\partial_v A_w) \vec{I} + A_w \partial_v \vec{I}$, and antisymmetrizing in v, w gives $\partial_v A_w - \partial_w A_v + [A_w, A_v] = 0$.) Flatness is a stringent consistency condition on any derived or transformed system, since it interlocks all matrix entries; it is preserved by every change of basis; and it guarantees that solutions continued along a path depend only on the endpoint and on the homotopy class of the path in the complement of the singular loci, which underlies Section 7.

3 Families, blocks, and equivalence

The double-real channel does not present one large differential system but many small ones, and many of the small ones coincide as differential systems once their bases are matched. Two equivalence relations — one among integral families, one among diagonal blocks — detect the coincidences and together set the true size of the analytic problem. Both are decided by exact identities: a change of loop momenta for families, a conjugation of matrices for blocks.

3.1 Equivalence of integral families

Families are generated mechanically, one propagator set per diagram, and diagrams proliferate: the channel begins with 1898 of them. Most are redundant, because different routings of the loop momenta describe the same set of denominators.

Definition 3.1 (Equivalent families; representative families). *Two integral families with L loop momenta are equivalent if an affine change of the loop momenta — a linear map composed with a shift, here the matrix M composed with a shift by external momenta —*

$$k_\ell \mapsto \sum_{m=1}^L M_{\ell m} k_m + \sum_r c_{\ell r} p_r, \quad M \in \text{GL}(L, \mathbb{Z}), \quad c_{\ell r} \in \mathbb{Q}, \quad (24)$$

maps one propagator set bijectively onto the other and maps cut lines to cut lines. A matrix in $\text{GL}(L, \mathbb{Z})$ has $|\det M| = 1$, so the integration measure is invariant and equivalent families have identical integrals up to a relabeling of the indices a_i . A chosen representative of each equivalence class is called a representative family.

Example 3.2 (Two routings of the one-mass triangle). *Let $p_1^2 = p_2^2 = 0$ and $(p_1 + p_2)^2 = s$, and consider the two one-loop families*

$$\mathcal{F}: \{k^2, (k + p_1)^2, (k + p_1 + p_2)^2\}, \quad \tilde{\mathcal{F}}: \{k^2, (k + p_1 + p_2)^2, (k + p_2)^2\}, \quad (25)$$

the same triangle graph with its external legs attached in a different order. No relabeling of the D_i maps one set onto the other. But the substitution $k \mapsto -k - p_1 - p_2$, of the form (24) with $M = (-1)$, maps $\tilde{\mathcal{F}}$ onto $\mathcal{F}$:

$$k^2 \mapsto (k + p_1 + p_2)^2, \quad (k + p_1 + p_2)^2 \mapsto k^2, \quad (k + p_2)^2 \mapsto (k + p_1)^2, \quad (26)$$

a bijection of the propagator sets. The families are equivalent, and every integral of one is an integral of the other with permuted indices.

Searching over the pairs (M, c) directly is not how equivalence is found: the search space is large, and the failure of a search proves nothing. Instead one computes an invariant that does not depend on the routing at all. For positive powers, the integral (11) has the Feynman-parametric representation

$$I_{a_1 \dots a_N} = (-1)^a \frac{\Gamma(a - Ld/2)}{\prod_i \Gamma(a_i)} \int_{x_i \geq 0} \left(\prod_i dx_i x_i^{a_i-1} \right) \delta\left(1 - \sum_i x_i\right) \frac{U^{a-(L+1)d/2}}{F^{a-Ld/2}}, \quad a = \sum_i a_i, \quad (27)$$

where U and F are the *Symanzik polynomials* of the graph,

$$U = \sum_T \prod_{e \notin T} x_e, \quad F = - \sum_{(T_1, T_2)} s_{(T_1, T_2)} \prod_{e \notin T_1 \cup T_2} x_e \quad (\text{massless lines}): \quad (28)$$

T runs over the spanning trees of the graph, (T_1, T_2) over its spanning two-forests, and $s_{(T_1, T_2)}$ is the square of the total external momentum flowing into T_1 . At one loop this reduces to $F = - \sum_{i < j} q_{ij}^2 x_i x_j$ with q_{ij} the difference of the momenta carried by lines i and j . For the bubble of Example 2.1 ($L = 1$, $a_1 = a_2 = 1$, $U = x_1 + x_2$, $F = -s x_1 x_2$), the representation (27) restricted to the simplex $x_1 + x_2 = 1$ gives

$$B = \Gamma(\varepsilon) (-s)^{-\varepsilon} \int_0^1 dx [x(1-x)]^{-\varepsilon} = \frac{\Gamma(\varepsilon) \Gamma(1-\varepsilon)^2}{\Gamma(2-2\varepsilon)} (-s)^{-\varepsilon} = c(\varepsilon) (-s)^{-\varepsilon}, \quad (29)$$

using $\Gamma(1+\varepsilon) = \varepsilon \Gamma(\varepsilon)$ and $\Gamma(2-2\varepsilon) = (1-2\varepsilon)\Gamma(1-2\varepsilon)$ in the last step: the normalization of (27), including its sign, and the constant of Example 2.1 confirm each other. The essential point is visible in (28): U and F are built from the graph and from the assignment of external momenta to its vertices — the loop momenta have disappeared. A substitution (24) therefore acts on the parametric representation as a mere permutation of the Feynman parameters, and so *equivalent families have identical Symanzik pairs (U, F) up to a permutation of the parameters.*

Example 3.3 (The triangle, in parameters). *For the two routings of Example 3.2,*

$$U = x_1 + x_2 + x_3 \text{ for both,} \quad F_{\mathcal{F}} = -s x_1 x_3, \quad F_{\tilde{\mathcal{F}}} = -s x_1 x_2, \quad (30)$$

by (28) or by the one-loop formula ($q_{13} = p_1 + p_2$ for $\mathcal{F}$, $q_{12} = p_1 + p_2$ for $\tilde{\mathcal{F}}$, all other q_{ij}^2 vanishing): the pairs coincide after $x_2 \leftrightarrow x_3$. The shift of Example 3.2 acts on the parameters as the relabeling $\tilde{x}_1 \mapsto x_3$, $\tilde{x}_2 \mapsto x_1$, $\tilde{x}_3 \mapsto x_2$ dictated by its action on the propagators, under which $-s \tilde{x}_1 \tilde{x}_2 \mapsto -s x_3 x_1$ as well. The two matchings differ by $x_1 \leftrightarrow x_3$, a symmetry of the pair $(x_1 + x_2 + x_3, -s x_1 x_3)$; matchings of the polynomials always form a coset of that symmetry group, and each member is a candidate for the underlying momentum shift. They are found on the polynomials, without any search over momenta.

Comparing polynomial pairs up to permutation is itself a finite problem, and Pak's algorithm [9] settles it canonically. Write the pair as a matrix with one row per monomial: the entries of a row are the exponents of the parameters in that monomial, followed by the coefficient, with the rows of U and of F kept in separate groups. For the triangle of Example 3.3 the matrix is

	x_1	x_2	x_3	coeff.	
U	1	0	0	1	
U	0	1	0	1	for $\mathcal{F}$,
U	0	0	1	1	and the row (1, 1, 0) in place of (1, 0, 1) for $\tilde{\mathcal{F}}$.
F	1	0	1	$-s$	(31)

A relabeling of the parameters permutes the columns; the rows are then sorted, and one seeks the relabeling making the sorted matrix lexicographically maximal. The search is greedy and refines column by column: at step j each surviving partial relabeling is extended by each parameter not yet placed, the resulting matrices are compared on their first j columns, and only the maximizers survive; the relabelings that survive to the end form a coset of the symmetry group of the pair, and are that group when the pair is already normally ordered. Here U is symmetric and only the F row decides: among (1, 1, 0), (1, 0, 1) and (0, 1, 1) the maximum is (1, 1, 0), so both routings normalize to

$$(x_1 + x_2 + x_3, -s x_1 x_2), \quad (32)$$

reached from $\tilde{\mathcal{F}}$ by the identity and from $\mathcal{F}$ by $x_2 \leftrightarrow x_3$, with the residual symmetry $x_1 \leftrightarrow x_2$ left over. The outcome is a *normal ordering* of the parameters, the same for every member of a permutation orbit, so two families are candidates for equivalence exactly when their normally ordered pairs are identical. Cut lines are carried through with distinguishing marks, so that the normal ordering can never trade a cut line for an uncut one; for a cut family the parametric representation serves only as this fingerprint of the underlying graph, never for evaluation.

A coincidence of normal orderings is a reliable *candidate* for equivalence; every candidate is then confirmed by constructing the substitution (24) explicitly. The permutation σ of parameters dictates which propagator must map onto which, and the requirement

$$\tilde{D}_i(Mk + c \cdot p) = D_{\sigma(i)}(k) \quad \text{for every } i \quad (33)$$

is solved as follows: writing each inverse propagator as the square of a linear form, (33) equates the two linear forms up to sign, one vector equation per line. The system is linear in (M, c) once a sign has been chosen for each line; the sign choices are a finite branching, and at one loop they are trivial, the sign being absorbed into M . There, with $D_j = (k + r_j)^2$ and $\tilde{D}_i = (k + \tilde{r}_i)^2$,

the conditions read $M(c \cdot p + \tilde{r}_i) = r_{\sigma(i)}$; for Example 3.2, $M = -1$ and $c \cdot p = -(p_1 + p_2)$ solve $p_1 + p_2 - \tilde{r}_i = r_{\sigma(i)}$ for the three lines at once. Applying the substitution so obtained is what turns the identification into a proof. The 1898 raw families of the channel collapse to 430 inequivalent ones, and only a fraction of the survivors supports master integrals at all: ninety-one families, in the end, carry the Pfaffian systems of the problem.

Differentiation is family-local — the derivative of a master of a family reduces to masters of that family and its subsectors, never to unrelated ones — so the differential-equation problem separates into 91 Pfaffian systems, of dimensions between 2 and 47, accounting for all 347 masters. (The dimensions sum to more than 347: each system contains, besides its own masters, the subsector masters it couples to, and those are counted once, in the family that owns them.) The systems are separate but not independent: a subsector master shared by two of them is one and the same function, so they are solved in an order compatible with the subsector relation, as Section 8 uses. The boundaries of the systems were fixed by the reduction itself; what must be constructed anew is, for each family, the connection (A_v, A_w) , together with the proof of its flatness.

3.2 Blocks and their equivalence

Definition 3.4 (Block). *Form the directed graph whose vertices are the masters of a system and which has an edge $i \rightarrow j$ whenever $A_{ji} \neq 0$, that is, whenever I_i occurs in the derivative of I_j . A block is a strongly connected component of this graph: a set of masters in which every two are joined in both directions by chains of such couplings. With the masters ordered by sector, the blocks are the diagonal entries a_{ss} of Eq. (22), and a block of size one is a master whose derivative involves only itself and lower sectors.*

Definition 3.5 (Equivalence of blocks; classes). *Let $b^{(1)}$ and $b^{(2)}$ be the connections of two blocks, in the same family or in different ones. The blocks are equivalent if there is a permutation matrix P with*

$$b^{(2)}(v, w) = P^{-1} b^{(1)}(v, w) P \quad \text{or} \quad b^{(2)}(v, w) = P^{-1} b^{(1)}(w, v) P, \quad (34)$$

as an identity of rational matrices. An equivalence class of blocks is called a class; each class is represented by a single block, whose canonical form is constructed once and serves for all members.

Because P is constant, the conjugation in (34) carries no derivative term ($P^{-1}dP = 0$): the equivalence is a plain identity of rational matrices, decided entry by entry. In the double-real channel the 91 family Pfaffian systems contain 1119 blocks, and the 1119 blocks fall into 173 classes.

Example 3.6 (Two equivalent blocks). *Suppose one family contains the 2×2 block*

$$b^{(1)} = \varepsilon \left[\begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix} d \log w + \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} d \log(1 - w) \right] \quad (35)$$

and another family the block

$$b^{(2)} = \varepsilon \left[\begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix} d \log v + \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} d \log(1 - v) \right]. \quad (36)$$

With $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ (reversal of the basis order) and the swap $v \leftrightarrow w$, conjugation sends $\begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix} \mapsto \begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix}$ and $\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \mapsto \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, so the second identity in (34) holds, and any canonical solution of $b^{(1)}$ becomes one of $b^{(2)}$ after the swap and the reversal. The block $b^{(1)}$ is the system whose solution will be constructed in Example 4.2, where its second component turns out to be the dilogarithm.

The roughly sixfold collapse from blocks to classes has a physical origin. A block is the differential system of the masters of one subtopology, and the same subtopology occurs as a subdiagram of many different parent diagrams: the identification of *complete* families by momentum shifts leaves untouched the far more frequent coincidence of their *subsectors*. The involution $v \leftrightarrow w$ is the exchange $p_1 \leftrightarrow p_2$ of the incoming partons, that is $t \leftrightarrow u$ — a relabeling of the initial state, not a crossing — which maps cut diagrams into cut diagrams. Equivalence is never assumed on such grounds, however: a candidate identification is accepted only when (34) is verified identically.

The analysis of Sections 5 and 6 is therefore carried out once per class rather than once per block, and each family is afterwards reconstructed from the canonical forms of its classes. Which rows of a family realize which class, and under which permutation of the basis and which exchange of the variables, is part of the statement being proved and is verified again at the reconstruction.

4 The canonical form

4.1 The ε -form and its immediate consequences

Definition 4.1 (Canonical form; letters, alphabet, residues). *A Pfaffian system is in canonical form (equivalently, in ε -form; both names are standard, after Henn [3]) if its connection is*

$$d\vec{F} = \varepsilon \sum_a R_a d \log \varphi_a \vec{F}, \quad (37)$$

where the residue matrices R_a are constant — independent of kinematics and regulator alike — and the letters φ_a are functions of the kinematics alone. The collection $\{\varphi_a\}$ is the alphabet of the system. In the simplest and most common case the letters are polynomials; Section 6 enlarges the notion to algebraic functions.

One family of the double-real channel, a system of dimension fourteen, carries the six-letter alphabet

$$\{v, w, 1-v, 1-w, v+w, 1-v-w\}, \quad (38)$$

and every entry of its canonical connection is ε times a $\mathbb{Q}$ -linear combination of logarithmic differentials of these six polynomials. Richer systems have richer alphabets — among the two-root families one finds letters quadratic in the variables — but always of the same form.

The equivalence of blocks, Definition 3.5, can now be stated on the data of (37): for two blocks in canonical form the identity (34) reduces to a statement about the residue matrices alone,

$$R_a^{(2)} = P^{-1} R_{\sigma(a)}^{(1)} P, \quad (39)$$

where σ is the relabeling of the letters induced by the exchange (under $v \leftrightarrow w$ the letter v becomes w , while $1-v-w$ is fixed, and so on). A class is thus a finite set of constant matrices up to simultaneous conjugation and relabeling.

Three consequences follow at once, and they are linked. The connection has simple poles exactly on the loci $\varphi_a = 0$ and nowhere else in the finite plane — the system is *Fuchsian*, and for generic ε each letter with non-vanishing residue is a genuine branch locus of the solutions. (Infinity generally carries a pole as well; Section 5.1 makes use of it.) Every solution is therefore analytic on the complement of $\bigcup_a \{\varphi_a = 0\}$ and has all its finite branch points among the zeros of letters: the alphabet is the complete list of singular loci. Second, because each $d \log \varphi_a$ is a

closed form, the flatness condition (23) becomes a condition on the commutators of the constant matrices, weighted by rational functions,

$$\sum_{a < b} [R_a, R_b] W_{ab}(v, w) = 0, \quad W_{ab} = \frac{\partial_v \varphi_a \partial_w \varphi_b - \partial_w \varphi_a \partial_v \varphi_b}{\varphi_a \varphi_b}, \quad (40)$$

where $d \log \varphi_a \wedge d \log \varphi_b = W_{ab} dv \wedge dw$. The W_{ab} are rational, not independent, and the independent conditions on the commutators are obtained by decomposing them in partial fractions and equating the coefficient of each independent pole structure to zero; that done, flatness is a finite list of commutator identities, checked once and for all. Third, as the next subsection shows, the coefficient of ε^k in the solution is a $\mathbb{Q}$ -linear combination of k -fold iterated integrals contracted with the boundary vector, so the expansion is homogeneous in *transcendental weight* — the grading on functions and numbers under which a logarithm carries weight one, a k -fold iterated integral of logarithmic forms weight k , and the constants ζ_k and π^k weight k , with rational numbers at weight zero. In the canonical basis this grading is manifest, order by order in ε .

4.2 Order-by-order integration

In a generic basis the regulator is entangled with the kinematics in every entry of A , and the coefficients of the ε -expansion obey coupled differential equations with no distinguished structure. In the canonical basis the regulator counts the number of integrations and does nothing else, and the system reduces to quadratures.

To see this, write $\hat{A} = \sum_a R_a d \log \varphi_a$, so that $d\vec{F} = \varepsilon \hat{A} \vec{F}$ with $\hat{A}$ independent of ε . Substituting $\vec{F} = \sum_k \varepsilon^k \vec{F}^{(k)}$ and comparing powers of ε :

$$d\vec{F}^{(0)} = 0, \quad d\vec{F}^{(k)} = \hat{A} \vec{F}^{(k-1)} \quad (k \geq 1). \quad (41)$$

The system has disentangled into a recursion of quadratures: the lowest order is a constant vector, and each subsequent order is an integral of known functions. Along a path γ from a base point p_0 ,

$$\vec{F}^{(k)}(p) = \vec{F}^{(k)}(p_0) + \int_{\gamma} \hat{A} \vec{F}^{(k-1)}, \quad (42)$$

and unwinding the recursion expresses the full solution as the path-ordered exponential

$$\vec{F}(p) = \mathbb{P} \exp \left(\varepsilon \int_{\gamma} \hat{A} \right) \vec{F}(p_0) = \sum_{k \geq 0} \varepsilon^k \sum_{a_1, \dots, a_k} R_{a_k} \cdots R_{a_1} \int_{0 \leq t_1 \leq \dots \leq t_k \leq 1} \prod_{i=1}^k d \log \varphi_{a_i}(\gamma(t_i)) \vec{F}(p_0). \quad (43)$$

The k -fold integrals are *Chen iterated integrals* [11] over the alphabet; flatness makes them independent of the choice of path within its homotopy class. The analytic problem is thereby solved in principle, and what remains is to express the iterated integrals in a standard basis of functions (Section 7) and to determine the boundary vector $\vec{F}(p_0)$ (Section 8).

The recursion (41) starts from a *constant* at order ε^0 , while the masters themselves have poles in ε . There is no contradiction: the matrix T relating the derived masters to the canonical ones carries inverse powers of ε from the normalization of the basis, so a finite canonical order feeds several Laurent orders of a master, and statements about uniform weight refer to the canonical basis.

In one dimension $dF = \varepsilon d \log w F$ has the solution $F = C w^\varepsilon = C(1 + \varepsilon \log w + \frac{\varepsilon^2}{2} \log^2 w + \dots)$, and the recursion (41) generates exactly that expansion. Both descriptions of this object — the

expanded logarithmic series and the unexpanded power w^ε — will be needed, and the tension between them is the subject of Section 9.

Example 4.2 (A two-dimensional system and the dilogarithm). *Take the alphabet $\{w, 1-w\}$ on $0 < w < 1$ and the system*

$$d\vec{F} = \varepsilon \left[\begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix} d \log w + \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} d \log(1-w) \right] \vec{F}, \quad (44)$$

with the boundary condition $\vec{F} \rightarrow (1, 0)^\top$ as $w \rightarrow 0$ (the precise sense in which a value is assigned at a singular point is given in Section 4.3). The first component stays constant. For the second, the recursion gives

$$F_2^{(1)}(w) = \int_0^w d \log(1-w') \cdot 1 = \log(1-w), \quad (45)$$

$$F_2^{(2)}(w) = \int_0^w \left[d \log(1-w') \cdot 0 - d \log w' \cdot F_2^{(1)}(w') \right] = - \int_0^w \frac{dw'}{w'} \log(1-w') = \text{Li}_2(w). \quad (46)$$

The dilogarithm appears as the twofold iterated integral with the word $(w, 1-w)$; the next order produces trilogarithms, and so on. Every master of the calculation is, order by order, a $\mathbb{Q}$ -linear combination of such words in its own alphabet.

4.3 Local behaviour at the singular loci

The canonical form does more than integrate: it displays, before any integration, how every solution behaves at every singular locus. Near a zero of a letter — say $w \rightarrow 0$, with letter w — the system reads

$$\partial_w \vec{F} = \frac{\varepsilon R_w}{w} \vec{F} + (\text{regular}), \quad (47)$$

and the classical Frobenius analysis applies: for generic ε every solution is

$$\vec{F} = H(w) w^{\varepsilon R_w} \vec{c} = \sum_{\text{modes}} w^{n+a\varepsilon} (\log w)^j (\text{const} + O(w)), \quad (48)$$

with H analytic at the origin and $H(0) = 1$, $\vec{c}$ a constant vector, and the powers $w^{\varepsilon R_w} = e^{\varepsilon R_w \log w}$ giving the modes on the right: the coefficients a range over the eigenvalues of R_w , non-negative integer shifts n from the Taylor expansion of the regular part, and powers of $\log w$ from the Jordan structure of R_w . *The complete local behaviour of every solution at every singular locus is thus encoded in the residue matrix at the corresponding letter* — exponents in its eigenvalues, logarithms in its Jordan blocks — and is known exactly in advance. Matching the global solution onto (48) is also what assigns boundary data at a singular point: the data are the components of $\vec{c}$, that is the coefficients of the modes in the Frobenius decomposition, of which the coefficient of the mode with exponent zero is the ordinary value when the solution is regular there.

Two qualifications keep the statement honest. The normal form requires that no two exponents $\varepsilon \lambda_i, \varepsilon \lambda_j$ differ by a nonzero integer — automatic for generic ε , which is the regime of the Laurent expansion, but not an identity in ε . And “logarithms come from the Jordan blocks” refers to the solution at fixed ε ; upon expanding in ε the logarithms proliferate for a second, independent reason,

$$w^{\varepsilon R_w} = e^{\varepsilon R_w \log w} = \sum_{m \geq 0} \frac{(\varepsilon \log w)^m}{m!} R_w^m, \quad (49)$$

so a power of $\log w$ appears at every order in ε whether or not R_w is diagonalizable; the Jordan structure bounds only the logarithms that survive at fixed ε . This distinction — the unexpanded mode versus its logarithmic series — is precisely what the endpoint analysis of Section 9 turns on.

Remark 4.3 (Existence). *No theorem guarantees that a given block admits a canonical form with rational letters, and a system with an irregular singularity admits none at all (Example 5.2). The canonical form is accordingly treated as a statement to be verified: any means whatsoever may produce the transformation, and what establishes it is the exact identity between the transformed connection and the form (37).*

In the double-real channel every class admits a canonical form, in a number of cases only after a change of variables that rationalizes a square root; for the families whose alphabets carry three independent roots no such change of variables exists, and Section 6 describes what takes its place.

5 Construction of the canonical transformation

The object to be constructed is an invertible matrix $T(v, w; \varepsilon)$ relating the derived basis to the canonical one, $\vec{I} = T \vec{F}$. Substituting into the Pfaffian system (15) shows that the connection transforms inhomogeneously,

$$A \mapsto A' = T^{-1}AT - T^{-1}dT, \quad (50)$$

and the requirement is that A' be of the form (37). The triangular structure (22) decides the order in which the blocks are treated: a diagonal block never feels what lies below it, so each is brought to canonical form on its own, once for each class, after which the entries below the diagonal yield to transformations of unipotent shape. What survives both steps is a dependence on ε constant in the kinematics, removed last by conjugation.

5.1 Diagonal blocks: Fuchsian reduction and balances

For the diagonal blocks two complementary strategies are classical, and each produces a candidate transformation which then stands or falls on the exact identity.

The first is the *rational ansatz*. Multiplying (50) by T from the left removes the inverse and leaves an identity between rational one-forms,

$$dT = AT - \varepsilon T \sum_a R_a d \log \varphi_a. \quad (51)$$

The candidate alphabet is not guessed: the letters must account for the singularities the system already has, so one takes the irreducible factors of the denominators of A , together with the irreducible factors of the discriminants that enter the eigenvalues of its residue matrices — the latter being where algebraic letters, if any, come from. One then posits for T a matrix of rational functions whose denominators are products of these letters and whose numerators are polynomials of bounded degree with undetermined ε -dependent coefficients. Clearing denominators and comparing coefficients of the independent monomials turns (51) into a finite algebraic system in the unknowns. It is bilinear — the products TR_a pair the numerator coefficients with the residues — so it is not one linear solve: one fixes the residues from a finite list of candidates and solves linearly for T in each case, or eliminates the residues first. With that caveat the ansatz is exhaustive at a given degree and a given alphabet: if the system has no solution, no transformation of that degree exists over that alphabet.

The second strategy proceeds through normal forms, in two steps: a higher pole is first reduced, if it can be, to a simple one, and the residue eigenvalues are then normalized. The first step is Moser's algorithm [6]: a rank condition on the leading coefficient of the pole decides whether a rational transformation lowering the Moser rank exists and constructs one when it does; a pole that the criterion cannot lower is a genuine irregular singularity, and then no canonical form exists at all (Example 5.2). The second step uses *balance transformations* between a pair of singular points v_1, v_2 ,

$$T = \bar{P} + \frac{v - v_1}{v - v_2} P, \quad \bar{P} = 1 - P, \quad (52)$$

with P a projector of rank one. Which projector is not free. Writing $A_{(1)}, A_{(2)}$ for the residues at v_1, v_2 and using $T^{-1} = \bar{P} + f^{-1}P$ with $f = (v - v_1)/(v - v_2)$,

$$T^{-1}AT = \bar{P}A\bar{P} + f\bar{P}AP + f^{-1}PA\bar{P} + PAP, \quad (53)$$

and the third term has a double pole at v_1 , the second a double pole at v_2 , unless

$$PA_{(1)}\bar{P} = 0 \quad \text{and} \quad \bar{P}A_{(2)}P = 0. \quad (54)$$

Both hold if one takes a *left* eigenvector η of $A_{(1)}$, with eigenvalue λ , and a *right* eigenvector u of $A_{(2)}$, with eigenvalue μ , such that $\eta^\top u \neq 0$, and sets

$$P = \frac{u\eta^\top}{\eta^\top u}. \quad (55)$$

A spectral projector of the residue at v_1 alone will not do: it satisfies the first condition and in general violates the second. For $A_{(1)} = \text{diag}(1, 0)$, $A_{(2)} = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}$ and $P = \text{diag}(1, 0)$ one finds $\bar{P}A_{(2)}P = \begin{pmatrix} 0 & 0 \\ 2 & 0 \end{pmatrix} \neq 0$, and the transformation creates a double pole at v_2 . With P as above the balance sends $\lambda \mapsto \lambda - 1$ and $\mu \mapsto \mu + 1$. It leaves the exponents at every other singular point unchanged — the residue at v_k is conjugated by the constant matrix $T(v_k)$ — and the residue at infinity untouched, since $T(\infty) = 1$. A balance necessarily acts at two points and never at one: for a Fuchsian connection $A = \sum_k A_k dv/(v - v_k)$ the residue at infinity is $-\sum_k A_k$, so the sum of all residues, that point included, vanishes identically and no transformation can change an exponent at a single point.

The target of the sequence is that every residue eigenvalue, which for these systems has the form $n + a\varepsilon$ with $n \in \mathbb{Z}$, be normalized to $a\varepsilon$: the integer parts must all be removed. A balance is admissible when λ has integer part $n_1 > 0$ at v_1 and μ has integer part $n_2 < 0$ at v_2 , and eigenvectors with $\eta^\top u \neq 0$ exist; it then decreases the non-negative integer $\sum |n|$, summed over all residue eigenvalues at all singular points including infinity, by two. The sequence therefore terminates. What can fail is the existence of an admissible pair at some step — the partner with an integer part of the opposite sign is guaranteed only in total, by the vanishing of the sum of all residues, not at a point where the eigenvector condition also holds.

Example 5.1 (A balance and the constant step). *Take the one-variable Fuchsian system with singular points 0 and 1,*

$$A = \frac{dv}{v} \begin{pmatrix} \varepsilon & 0 \\ 0 & 1 + \varepsilon \end{pmatrix} + \frac{dv}{v - 1} \begin{pmatrix} \varepsilon & 0 \\ c & -1 + \varepsilon \end{pmatrix}. \quad (56)$$

The exponent $1 + \varepsilon$ at $v = 0$ is too high by one and the exponent $-1 + \varepsilon$ at $v = 1$ too low by one: an admissible pair. A left eigenvector of the first residue for $1 + \varepsilon$ is $\eta^\top = (0, 1)$; a right eigenvector

of the second for $-1 + \varepsilon$ is $u = (0, 1)^\top$, since $(A_{(2)} - (-1 + \varepsilon))u = \begin{pmatrix} 1 & 0 \\ c & 0 \end{pmatrix} u = 0$; and $\eta^\top u = 1 \neq 0$. Hence $P = \text{diag}(0, 1)$, both conditions (54) hold because the products are diagonal or vanish, and (52) is $T = \text{diag}(1, v/(v-1))$. A diagonal T acts by $A'_{ij} = A_{ij} t_j/t_i$ off the diagonal and subtracts $d \log t_i$ on it, so

$$A' = \frac{dv}{v} \begin{pmatrix} \varepsilon & 0 \\ c & \varepsilon \end{pmatrix} + \frac{dv}{v-1} \begin{pmatrix} \varepsilon & 0 \\ 0 & \varepsilon \end{pmatrix}, \quad (57)$$

using $\frac{1+\varepsilon}{v} + \frac{-1+\varepsilon}{v-1} - \left(\frac{1}{v} - \frac{1}{v-1}\right) = \frac{\varepsilon}{v} + \frac{\varepsilon}{v-1}$ for the second diagonal entry and $\frac{c}{v-1} \cdot \frac{v-1}{v} = \frac{c}{v}$ for the lower left. All exponents are now ε , but the entry c is not proportional to it. The constant transformation $T_c = \text{diag}(\varepsilon, 1)$ completes the reduction:

$$A'' = T_c^{-1} A' T_c = \varepsilon \left[\begin{pmatrix} 1 & 0 \\ c & 1 \end{pmatrix} d \log v + \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} d \log(1-v) \right], \quad (58)$$

the canonical form, with constant residues and the alphabet $\{v, 1-v\}$.

Example 5.2 (Irregular and removable higher poles). Consider first the scalar equation

$$\frac{dF}{dw} = \left(\frac{\alpha}{w^2} + \frac{\beta \varepsilon}{w} \right) F. \quad (59)$$

The substitution $F = e^{-\alpha/w} G$ does remove the double pole, since $\frac{d}{dw} e^{-\alpha/w} = \frac{\alpha}{w^2} e^{-\alpha/w}$, leaving $G' = (\beta \varepsilon/w) G$ and $G = C w^{\beta \varepsilon}$. But $e^{-\alpha/w}$ is not a rational — not even a meromorphic — change of basis, and no rational one can do the job: a rational $F = r(w) G$ shifts the connection by $-d \log r$, which has only simple poles. A scalar double pole is therefore a genuinely irregular singularity, and no canonical form exists.

What is removable is a higher pole sitting in a nilpotent direction of a matrix residue. For

$$A = \frac{1}{w^2} \begin{pmatrix} 0 & \alpha \\ 0 & 0 \end{pmatrix} dw, \quad T = \begin{pmatrix} 1 & 0 \\ 0 & w \end{pmatrix} \implies A' = \frac{1}{w} \begin{pmatrix} 0 & \alpha \\ 0 & -1 \end{pmatrix} dw, \quad (60)$$

a simple pole, reached by a rational transformation — Fuchsian, and ready for the balances above. In the present calculation the genuinely irregular case never occurs once the variables of Section 6 are chosen correctly.

5.2 The second variable

Moser reduction and balances are constructions for an ordinary differential equation in one variable; the systems here have two. The way through is to treat the variables in turn, and it is the second step, not the first, that decides whether a canonical form is reached.

Fix w as a symbol and read the v -component of the Pfaffian system,

$$\partial_v \vec{I} = A_v(v, w; \varepsilon) \vec{I}, \quad (61)$$

as an ordinary differential equation in v over the field $\mathbb{Q}(w, \varepsilon)$ — or over a finite algebraic extension of it, since the singular points of this equation are the v -roots of the letters, which are algebraic functions of w : for the letter $1 - v - w$ the root is $v = 1 - w$, for $1 - 4vw$ it is $v = 1/(4w)$, and for a letter quadratic in v they are a conjugate pair. Everything of Section 5.1 applies verbatim over that field, the spectator w being carried along symbolically, and produces a transformation $T_1(v, w; \varepsilon)$ with

$$T_1^{-1} A_v T_1 - T_1^{-1} \partial_v T_1 = \varepsilon \sum_a R_a \partial_v \log \varphi_a, \quad \partial_v R_a = 0. \quad (62)$$

The R_a here are independent of v — elements of $\mathbb{Q}(w, \varepsilon)$, not yet constants in the sense of the canonical-form definition; the residual dependence on the regulator is removed at the end, in Section 5.5. Nothing has been required of the w -direction, and in general

$$A'_w = T_1^{-1} A_w T_1 - T_1^{-1} \partial_w T_1 \quad (63)$$

is not canonical.

The freedom left is a v -independent gauge $U(w; \varepsilon)$. Since $\partial_v U = 0$, replacing T_1 by $T_1 U$ leaves the v -direction free of an inhomogeneous term and merely conjugates its residues,

$$A''_v = \varepsilon \sum_a (U^{-1} R_a U) \partial_v \log \varphi_a, \quad A''_w = U^{-1} A'_w U - U^{-1} \partial_w U, \quad (64)$$

while in the w -direction it acts in full. In the canonical form one and the same residue matrix stands in both directions, so two requirements fall on U . The conjugated residues must again be constant,

$$U^{-1} R_a U = R''_a, \quad R''_a \text{ independent of } w, \quad (65)$$

an algebraic condition; and the w -connection must be canonical with those same residues, which, written without the inverse, is

$$\partial_w U = A'_w U - \varepsilon U \sum_a R''_a \partial_w \log \varphi_a, \quad (66)$$

a *linear* equation for U once the R''_a are fixed. The pair (65), (66) is bilinear in (U, R''_a) , as the ansatz of Section 5.1 and the conjugation of Section 5.5 are in their unknowns, and is resolved the same way: once the R''_a are fixed from a finite list of candidates, both conditions are linear in U and the pair is one nullspace computation over $\mathbb{Q}(w, \varepsilon)$, of finite size, carried out once. Flatness of the original system is a necessary condition for a solution to exist and is not a sufficient one: whether the pair is solvable is decided by the linear solve itself, and nothing here asserts that the w -direction follows automatically from the v -direction.

Two features of the outcome are worth recording. The balances of Section 5.1 introduce apparent singularities in v at ε -dependent points; in the w -direction these appear as pure scalar $d \log$ terms with integer residues, and a rational scalar gauge removes them, so that they do not survive in the final alphabet. And the acceptance criterion is the two-variable identity: the composed transformation is admitted only when (37) holds in *both* directions with one and the same set of constant residues. A one-variable check establishes nothing about a two-variable system.

5.3 The entries below the diagonal: the completion equation

Suppose the diagonal blocks are already canonical — each equal to εe_k with e_k a constant-residue logarithmic form — while the entries below the diagonal, denoted b_{kj} with $j < k$, are not. They are brought to canonical form one block row at a time, by transformations of unipotent shape:

$$T = 1 + D, \quad D \text{ supported on block row } k, \text{ columns } j < k. \quad (67)$$

Because D is strictly triangular, $D^2 = 0$ and $T^{-1} = 1 - D$; moreover DAD and $D dD$ vanish — the latter because D and dD share the same support, the former because A is block lower triangular, so its (j, k) entries with $j < k$ are zero — and the transformation law (50) is exact without expansion:

$$A' = A + AD - DA - dD. \quad (68)$$

Reading off the (k, j) block of the requirement that A' be canonical gives the *completion equation*

$$dD_{kj} = \varepsilon(e_k D_{kj} - D_{kj} c_j) + \bar{b}_{kj} - \varepsilon \sum_a K_a d \log \varphi_a, \quad (69)$$

in which c_j is the canonical form of the diagonal block in column j (so that $a_{jj} = \varepsilon c_j$, in parallel with $a_{kk} = \varepsilon e_k$), the source $\bar{b}_{kj} = b_{kj} - \sum_{j < m < k} D_{km} b_{mj}$ collects the original coupling together with the contributions of blocks of the same row completed previously, and the matrices K_a — constant in the kinematics, though at this point still allowed to depend on the regulator, which Section 5.5 removes — are unknowns alongside D_{kj} : they become the residues of the completed entry. The essential feature of (69) is its structure in the unknowns (D_{kj}, K_a) : it is *affine-linear* — a linear operator applied to the unknowns, plus the fixed inhomogeneous source $\bar{b}_{kj}$. The standard consequence follows at once: its solution set is an affine subspace, one particular solution plus the null space of the linear part. That null space is exactly the non-uniqueness of Remark 5.5, and the same affine structure will appear once more, in the constraint system of Section 8.3.

Example 5.3 (A completion in one variable). *In one variable, with diagonal data $e = d \log v$ and $c = 0$, take the coupling*

$$b = \varepsilon dv + \frac{dv}{1+v}. \quad (70)$$

The two pieces are absorbed by different mechanisms. For the first, try a rational D : with $D = \frac{\varepsilon v}{1-\varepsilon}$ one has

$$\varepsilon e D + \varepsilon dv = \varepsilon \frac{dv}{v} \cdot \frac{\varepsilon v}{1-\varepsilon} + \varepsilon dv = \frac{\varepsilon}{1-\varepsilon} dv = dD, \quad (71)$$

so the non-logarithmic part of the coupling disappears into the transformation. The second piece is itself a logarithmic form with constant coefficient, $d \log(1+v)$: no rational D can absorb it (that would require $\log(1+v)$ itself), and none should — it is kept as a residue. Matching it in (69) requires $\varepsilon K_{1+v} = 1$, so the completed entry carries $d \log(1+v)$ at order ε^0 instead of at order ε . That mismatch is a normalization of the basis, constant in the kinematics, and is removed by the final transformation of Section 5.5. Rational parts of a coupling are absorbed by D ; logarithmic parts with constant coefficients become residues; normalization mismatches in ε are deferred to the end — the same three cases exhaust the general situation.

5.4 Solution of the completion equation by modular arithmetic

For the systems of interest the unknown D_{kj} may have several hundred rational-function entries of high degree, and solving (69) symbolically is prohibitive. Being affine-linear, it can instead be solved over finite fields and reassembled exactly. One writes $D_{kj} = N(v, w; \varepsilon) / \prod_a \varphi_a^{m_a}$, with the denominator a product of letters and the numerator a polynomial with undetermined coefficients. The ansatz is bounded by the source. Suppose $\bar{b}_{kj}$ has a pole of order p_a at $\varphi_a = 0$ while D_{kj} has one of order $m_a > p_a$, with leading coefficient $D^{(m)}$. At order $m_a + 1$ in that pole only two terms of (69) contribute, dD_{kj} and $\varepsilon(e_k D_{kj} - D_{kj} c_j)$, and they must cancel:

$$-m_a D^{(m)} = \varepsilon(R^e D^{(m)} - D^{(m)} R^c), \quad (72)$$

with R^e, R^c the residues of the diagonal data at that letter. Thus $D^{(m)}$ is an eigenvector of the linear map $X \mapsto \varepsilon(R^e X - X R^c)$ with eigenvalue $-m_a$; the eigenvalues of that map are $\varepsilon(\rho_i - \rho'_j)$

with ρ_i, ρ'_j the eigenvalues of R^e and R^c , and for generic ε none of them is a nonzero integer. Hence $D^{(m)} = 0$ and

$$m_a \leq p_a, \quad \deg N \leq \sum_a m_a \deg \varphi_a + \deg \bar{b}_{kj} + 1, \quad (73)$$

the second a degree count in the same spirit, with $\deg \bar{b}_{kj}$ the excess of the numerator degree of the source over its denominator degree. The degree bound is a bound and not a value: if the fit is inconsistent it is raised and the fit repeated, and inconsistency is exactly what an over-determined linear system reports. Substituting rational values of (v, w) and of the regulator, all reduced modulo a large prime p , turns (69) into exact linear equations over the finite field $\mathbb{F}_p$ for the unknown coefficients; enough substitutions determine them modulo p . The dependence on ε is recovered by the same device: the solution is a rational function of the regulator, and a rational function with numerator and denominator degrees d_N and d_D is determined by $d_N + d_D + 1$ samples — a polynomial of degree d by $d + 1$ — so one samples until the reconstruction stabilizes and withholds a further handful of regulator values as a check. Repeating over several primes, combining by the Chinese remainder theorem, and lifting each coefficient to a rational number by rational reconstruction yields an exact candidate, confirmed on evaluations withheld from the fit and at a fresh prime. None of this is a proof; the proof is the substitution of the reconstructed (D, K) back into (69) — or into the equivalent statement for the fully assembled family — and the exact verification of the identity.

Example 5.4 (Rational reconstruction). *Take $p = 101$ and suppose a coefficient is determined as $29 \bmod 101$. Rational reconstruction seeks a fraction of small numerator and denominator congruent to it: since $7 \cdot 29 = 203 = 2 \cdot 101 + 1 \equiv 1 \pmod{101}$, one has $29 \equiv \frac{1}{7} \pmod{101}$. A second prime, $p = 103$, where $\frac{1}{7} \equiv 59 \pmod{103}$ (as $7 \cdot 59 = 413 = 4 \cdot 103 + 1$), and the Chinese remainder lift of $(29, 59)$ modulo $101 \cdot 103$ reconstructs $\frac{1}{7}$ again. The guarantee behind the search is a lattice bound: modulo a single prime p the reconstruction is unique only for numerator and denominator below $\sqrt{p/2}$ — about 7.1 for $p = 101$, so $\frac{1}{7}$ is admissible there only just, which is exactly why the second prime is not optional.*

Remark 5.5 (Non-uniqueness of the completion). *When the linear part of (69) has a null space the completed entry is not unique: the affine solution set noted above is then a positive-dimensional affine subspace. Two solutions differ by an element of that null space, and the canonical forms they produce are related by a constant transformation that preserves the diagonal blocks: the two are equally canonical, but their residues K_a are not the same matrices. What the choice does affect materially is the degrees of the rational functions inserted into the sources $\bar{b}$ of the rows still to be completed, and degree growth compounds from row to row. A definite and favourable choice results from requiring the coefficients of the highest admissible numerator monomials to vanish, which selects a representative of low degree.*

5.5 The residual dependence on the regulator

After the completion, the connection is logarithmic with ε -independent letters, but its residue matrices may still depend on ε — rational functions of ε in place of the required constants times ε . The origin of this residue is the normalization of the master basis: each sector's masters were defined with ε -dependent prefactors, and a change of normalization is a transformation *constant in the kinematics*. For constant $T(\varepsilon)$ the inhomogeneous term of (50) vanishes and the transformation acts by simultaneous conjugation of all residues:

$$T^{-1}(\varepsilon) \tilde{R}_a(\varepsilon) T(\varepsilon) = \varepsilon R_a \quad \text{for all } a. \quad (74)$$

Written without the inverse, $\tilde{R}_a(\varepsilon) T(\varepsilon) = \varepsilon T(\varepsilon) R_a$, the condition is linear in T once the target residues are fixed — a finite-dimensional problem, solved at sampled rational values of the regulator and interpolated, with the cheap necessary test that the eigenvalues of $\tilde{R}_a(\varepsilon)/\varepsilon$ be ε -independent applied first. When a solution exists it is unique up to right multiplication by a matrix commuting with every R_a . Composing the class transformations, the row completions, and this constant conjugation yields the matrix T carrying the derived system (14) into the canonical form (37).

6 Algebraic letters: square roots and their rationalization

6.1 Square roots from the geometry of phase space

Nothing so far required the letters to be polynomials, and for a minority of the blocks they are not. The origin is the geometry of the momenta. Eliminating the direction of an unobserved momentum at fixed invariants requires solving quadratic equations, and the boundary of the region in which such an equation has real solutions is the vanishing of a Gram determinant. For a sub-configuration carrying three momenta $P_1, P_2, P_3 = -P_1 - P_2$ that Gram determinant is a Källén polynomial: with $2P_1 \cdot P_2 = P_3^2 - P_1^2 - P_2^2$,

$$\det \begin{pmatrix} 2P_1^2 & 2P_1 \cdot P_2 \\ 2P_1 \cdot P_2 & 2P_2^2 \end{pmatrix} = 4P_1^2 P_2^2 - (P_3^2 - P_1^2 - P_2^2)^2 = -\lambda(P_1^2, P_2^2, P_3^2), \quad (75)$$

with $\lambda(a, b, c) = a^2 + b^2 + c^2 - 2ab - 2bc - 2ca$ the triangle function, for which $\lambda(a, b, c) = (c - a - b)^2 - 4ab$. The sub-configurations of the double-real channel that produce roots carry the virtualities $s, -t, -u$, and λ is homogeneous of degree two, so that $\lambda(s, -t, -u)/s^2 = \lambda(1, v, w)$: dividing (75) by s^2 gives the polynomial that governs them,

$$\lambda_1(v, w) = \lambda(1, v, w) = (1 - v - w)^2 - 4vw. \quad (76)$$

Its zero locus lies *inside* the physical triangle (7) rather than on its boundary. Indeed

$$\lambda_1 = \left(1 - (\sqrt{v} + \sqrt{w})^2\right) \left(1 - (\sqrt{v} - \sqrt{w})^2\right), \quad (77)$$

and the second factor is positive throughout the triangle, so λ_1 vanishes exactly on $\sqrt{v} + \sqrt{w} = 1$ — through the interior point $v = w = \frac{1}{4}$ — and is positive on the side of it containing the origin, where $\sqrt{\lambda_1}$ is real. In the energy and angle of (8), with $\xi = v + w = 2E_3/\sqrt{s}$ and $4vw = \xi^2 \sin^2 \theta$, the locus is $\xi(1 + |\sin \theta|) = 1$ — an unremarkable configuration of the final state, at which the masters of the sectors sensitive to it nevertheless acquire a branch point of square-root type: a pseudo-threshold. A square-root branch point cannot be produced by logarithmic differentials of polynomials alone: the alphabet must contain $\sqrt{\lambda_1}$, typically through letters of the form

$$\frac{1 - v - w - \sqrt{\lambda_1}}{1 - v - w + \sqrt{\lambda_1}}, \quad (78)$$

which are exchanged with their reciprocals under the sign change $\sqrt{\lambda_1} \rightarrow -\sqrt{\lambda_1}$ (in the language of covering spaces, the deck transformation of the double cover). The double-real channel realizes six such radicands:

$$\lambda_1, \quad \lambda_2 = \lambda_1(-v, w), \quad \lambda_3 = \lambda_1(v, -w), \quad 4v + w^2, \quad v^2 + 4w, \quad 1 - 4vw. \quad (79)$$

Under the involution $v \leftrightarrow w$ — the exchange of the incoming partons, used in Definition 3.5 — the pairs $\{\lambda_2, \lambda_3\}$ and $\{4v + w^2, v^2 + 4w\}$ are exchanged, while λ_1 and $1 - 4vw$ are invariant; in the chart of Example 6.3 the involution reads $(x, y) \mapsto (1 - x, 1 - y)$.

Definition 6.1 (Root content of a system). *Radicands are counted modulo perfect-square factors and modulo products: $\sqrt{q_1}$, $\sqrt{q_2}$ and $\sqrt{q_1 q_2}$ constitute two independent roots, since the third is determined by the other two. Formally, the radicands of a system generate a subgroup of the square-class group $\mathbb{Q}(v, w)^\times / (\mathbb{Q}(v, w)^\times)^2$, and the root content is the rank of that subgroup.*

The rank is computed by elementary linear algebra over $\mathbb{F}_2$. Factor each radicand into irreducible polynomials, $q_i = \prod_\alpha \pi_\alpha^{n_{i\alpha}}$, and record only the parities: a radicand is a square exactly when every $n_{i\alpha}$ is even, and a product of radicands corresponds to the sum of their exponent vectors. The root content is therefore the rank over $\mathbb{F}_2$ of the matrix

$$(n_{i\alpha} \bmod 2)_{i\alpha}, \quad (80)$$

one row per radicand, one column per irreducible factor. For λ_1 , λ_2 , $\lambda_1 \lambda_2$ the third row is the sum of the first two, and the rank is two.

The radicands themselves are read off the derived connection, before any transformation of it: the discriminants of the irreducible quadratic factors of its singular loci, together with those occurring in the eigenvalues of its residue matrices. What this yields is an *upper bound* on the root content, not its value: a quadratic factor may be an apparent singularity, removable by a transformation, and its discriminant then contributes no root to the final alphabet. The bound is what selects a first candidate among the constructions below; a smaller root content is discovered when the construction succeeds without the corresponding root. In the double-real channel about half of the families involve no root at all; a third involve one; a dozen involve two; and three families — the hard case treated in Section 6.4 — involve three.

6.2 Rationalization by change of variables

Definition 6.2 (Rationalizing substitution). *A rationalizing substitution for a radicand $q(v, w)$ is a rational change of variables $(v, w) = (f(x, y), g(x, y))$ whose Jacobian determinant is not identically zero and for which $q(f, g) = r(x, y)^2$ with r a rational function. Under such a substitution $\sqrt{q}$ becomes the rational function r , and a system whose alphabet involves only $\sqrt{q}$ acquires polynomial letters in (x, y) .*

Example 6.3 (Rationalizing the Källén root). *Consider the substitution suggested by the factorized form of the degenerate configurations,*

$$v = xy, \quad w = (1 - x)(1 - y). \quad (81)$$

Then

$$1 - v - w = x + y - 2xy, \quad \lambda_1 = (x + y - 2xy)^2 - 4xy(1 - x)(1 - y) = (x - y)^2, \quad (82)$$

by direct expansion. Hence $\sqrt{\lambda_1} = x - y$, the Jacobian determinant is $\partial(v, w)/\partial(x, y) = x - y$, non-vanishing away from the diagonal, and the algebraic letter above becomes

$$\frac{(x + y - 2xy) - (x - y)}{(x + y - 2xy) + (x - y)} = \frac{y(1 - x)}{x(1 - y)}, \quad (83)$$

a ratio of polynomial letters. Two facts about the substitution deserve recording. It is two-to-one: both v and w are symmetric under $x \leftrightarrow y$, and that exchange is precisely the deck transformation $\sqrt{\lambda_1} \rightarrow -\sqrt{\lambda_1}$, branched on the diagonal $x = y$ where the Jacobian vanishes. And its real image is

not the whole triangle (7) but the sub-region $\sqrt{v} + \sqrt{w} \leq 1$, i.e. $\lambda_1 \geq 0$; outside it $\sqrt{\lambda_1}$ is imaginary, and reaching it is an analytic continuation to be performed deliberately. Paths chosen in (x, y) therefore live in that sub-region. The sign variants $v = \pm xy$, $w = \mp(1-x)(1-y)$ rationalize λ_2 and λ_3 in the same way, and substitutions of the same conic type rationalize $4v + w^2$, $v^2 + 4w$ and $1 - 4vw$ — one substitution for each of the six radicands.

The construction behind all such substitutions is classical: the equation $q(v, w) = r^2$ defines a conic in suitable projective coordinates, and a conic possessing one rational point is rationally parametrized by the pencil of lines through that point — the same device that parametrizes the Pythagorean triples. In the case needed here the rational point can always be taken at infinity. Regard q as a quadratic in one of the variables,

$$q = a(v)w^2 + b(v)w + c(v), \quad a = \alpha^2 \text{ a perfect square}, \quad (84)$$

and set $r = \alpha w + \tau$ with τ a new variable. The squares $\alpha^2 w^2$ cancel from $r^2 = q$, leaving a linear equation for w :

$$2\alpha w \tau + \tau^2 = bw + c \quad \implies \quad w = \frac{c - \tau^2}{2\alpha\tau - b}, \quad \sqrt{q} = \alpha w + \tau. \quad (85)$$

The pair (v, τ) is a rationalizing chart for q , and the identity $q = r^2$ in it is a matter of expansion. If the leading coefficient is not a square, one exchanges the roles of v and w , or shifts w so that the constant term c becomes a square and applies (85) to the reciprocal variable. The systematic treatment of such substitutions is that of Besier, van Straten, Wasser and Weinzierl [10]. The symmetric chart of Example 6.3 is an equivalent choice adapted to the involution $v \leftrightarrow w$: for λ_1 as a quadratic in w one has $\alpha = 1$, $b = -2(1 + v)$, $c = (1 - v)^2$, and (85) rationalizes it just as well.

Working in the new variables requires one transformation of the whole system by the chain rule,

$$A_x = A_v \partial_x f + A_w \partial_x g, \quad A_y = A_v \partial_y f + A_w \partial_y g, \quad (86)$$

with flatness re-established in (x, y) ; afterwards the constructions of Sections 4–5 apply verbatim.

6.3 Several roots

A system whose alphabet involves two independent roots requires a substitution rationalizing both at once, and such substitutions are found by iterating (85). Rationalize the first root; under that substitution the second radicand becomes a polynomial $q_2(x, y)$ in the new variables, and (85) applied to q_2 in one of them rationalizes it as well, leaving the first root — already rational — untouched. Two conditions must hold for the step to succeed, and neither is automatic. The pullback of a conic under a map of degree two is in general a quartic, so q_2 must still be of degree two in the variable chosen; and its leading coefficient in that variable must be a perfect square, so that the rational point at infinity used by (85) is available. The choice of variable is therefore not free, and for a general pair of radicands the construction stops here. For the Källén pairs of the double-real channel both conditions are met.

Example 6.4 (Two Källén roots at once). *Start from the chart of Example 6.3, in which $\sqrt{\lambda_1} = x - y$. In the same chart the second radicand $\lambda_2 = \lambda_1(-v, w) = (1 + v - w)^2 + 4vw$ becomes, using $1 + v - w = x + y$,*

$$\lambda_2 = (x + y)^2 + 4xy(1 - x)(1 - y) = (1 - 2x)^2 y^2 + 2x(3 - 2x)y + x^2, \quad (87)$$

a quadratic in y whose leading coefficient is the square $(1 - 2x)^2$. Its discriminant, $32x^2(1 - x)$, is not a square, so λ_2 does not factor into rational linear pieces; but that is not what is needed. Applying (85) with $\alpha = 1 - 2x$, $b = 2x(3 - 2x)$, $c = x^2$ and the new variable τ ,

$$y = \frac{x^2 - \tau^2}{2[(1 - 2x)\tau - x(3 - 2x)]}, \quad \sqrt{\lambda_2} = (1 - 2x)y + \tau, \quad (88)$$

and in the chart (x, τ) both roots are rational: $v = xy$, $w = (1 - x)(1 - y)$ and $\sqrt{\lambda_1} = x - y$ with y the rational function just displayed. The check is quick at $x = \frac{1}{2}$, where $\lambda_2 = 2y + \frac{1}{4}$ and the formula gives $y = \frac{1}{2}(\tau^2 - \frac{1}{4})$, so that $\lambda_2 = \tau^2$ indeed. Three substitutions of this kind — one for each pair of Källén-type radicands — cover every two-root family that occurs, and the canonical connections of those families are written in the final pair of parameters.

6.4 The Galois algebra of the roots

For the families involving three independent roots, no rational substitution rationalizes all three simultaneously. This follows from the geometry of the covering surface, not from a failure of search, and the argument is worth carrying out, since it is the only place in these notes where a construction is ruled out rather than performed.

A substitution rationalizing q_1, q_2, q_3 at once is a rational map from the plane onto the surface X on which all three roots are single-valued: the $(\mathbb{Z}/2)^3$ cover of the (v, w) -plane branched over the three conics $q_i = 0$. Adjoining the three roots as coordinates gives X a projective model in $\mathbb{P}^5$, in homogeneous coordinates (v, w, h, r_1, r_2, r_3) ,

$$r_i^2 = Q_i(v, w, h) = h^2 q_i(v/h, w/h), \quad i = 1, 2, 3, \quad (89)$$

with Q_i the degree-two homogenization of q_i , so that each equation is a genuine quadric. The model is a complete intersection of three quadrics, of degree $2^3 = 8$, matching the eight sheets of the cover. Its singularities sit over the points where two branch conics meet. The pairwise intersections are transverse except for finitely many simple tangencies, and each tangency lifts to two *ordinary double points* of the cover — points at which the surface looks locally like $xy = z^2$, the simplest surface singularity, called a node; in suitable local coordinates the lift is the equation $r_2^2 - r_1^2 + z^2 = 0$. How many there are is immaterial, because nodes are rational double points and their resolution is *crepant*: the canonical class pulls back without correction, $K_{\tilde{X}} = K_X$. For a complete intersection of three quadrics in $\mathbb{P}^5$ the adjunction formula gives $K = \mathcal{O}(\sum_i d_i - n - 1) = \mathcal{O}(6 - 6)$, trivial. A surface with trivial canonical class, vanishing irregularity $h^1(\tilde{X}, \mathcal{O}_{\tilde{X}}) = 0$ and geometric genus $p_g = \dim H^0(\tilde{X}, K_{\tilde{X}}) = 1$ is by definition a K3 surface, and $\tilde{X}$ has these invariants.

The conclusion follows from two classical facts, for which Beauville's book [13] is the reference. A surface is *unirational* if some rational map from the plane onto it is dominant — which is exactly what a simultaneous rationalizing substitution would provide. By Castelnuovo's theorem, in characteristic zero a unirational surface is rational; and a rational surface has $p_g = 0$, since p_g is a birational invariant and the plane has no holomorphic two-forms. But $\tilde{X}$ has $p_g = 1$. No such substitution exists.

The response is to renounce parametrization and compute in the extension the roots generate. Adjoin them as symbols:

$$E = \mathbb{Q}(v, w, \varepsilon) [r_1, r_2, r_3] / (r_i^2 - q_i), \quad (90)$$

an algebra of dimension 2^3 with basis $r_S = \prod_{i \in S} r_i$ indexed by subsets $S \subseteq \{1, 2, 3\}$; when the q_i are independent square classes — which is what root content three means — E is a field extension

of degree eight. Its Galois group is $(\mathbb{Z}/2)^3$, acting by the independent sign changes $r_i \mapsto -r_i$. Every element decomposes into the characters of this group (its *grades*); multiplication respects the grading,

$$r_S r_T = \left(\prod_{i \in S \cap T} q_i \right) r_{S \Delta T}, \quad (91)$$

and differentiation acts grade-wise through $dr_S = \frac{1}{2} r_S \sum_{i \in S} d \log q_i$. All linear problems of Section 5 — the completion equation, its evaluation over finite fields, the reconstruction — can be posed on the 8-component coefficient vectors, whose entries are ordinary rational functions. Evaluations modulo a prime are performed at points where every q_i is a nonzero quadratic residue, so that all eight embeddings of the algebra into $\mathbb{F}_p$ exist and the grade decomposition becomes a discrete Fourier transform over $(\mathbb{Z}/2)^3$; agreement of a solution across all eight sign embeddings is a consistency requirement that a correct computation must, and does, satisfy. No better description of these systems is presently known. What the construction requires of a system is one checkable property: that its alphabet involve three independent square classes drawn from the radicands (79), so that the grading is by $(\mathbb{Z}/2)^3$ and the rules above apply verbatim. The three families of the channel that have root content three have that property; carrying the construction through on them is part of the work that Section 10 records as remaining. It has so far been carried through on model systems of every root rank.

Example 6.5 (One root, in components). *The mechanism is already complete for $k = 1$. Let $E = \mathbb{Q}(v)[r]/(r^2 - q)$ with $q = q(v)$ not a square, so that $\{1, r\}$ is a basis and every element is $c_0 + c_1 r$ with c_0, c_1 rational. Multiplication in this basis is the matrix*

$$(c_0 + c_1 r)(z_0 + z_1 r) = (c_0 z_0 + c_1 z_1 q) + (c_0 z_1 + c_1 z_0) r, \quad \begin{pmatrix} z_0 \\ z_1 \end{pmatrix} \mapsto \begin{pmatrix} c_0 & c_1 q \\ c_1 & c_0 \end{pmatrix} \begin{pmatrix} z_0 \\ z_1 \end{pmatrix}, \quad (92)$$

whose determinant $c_0^2 - c_1^2 q$ is the norm; it vanishes only for $c_0 = c_1 = 0$, which is why E is a field. Differentiation is grade-preserving up to the factor r : from $r^2 = q$ one has $2r dr = dq$, that is

$$dr = \frac{1}{2} r d \log q. \quad (93)$$

Consequently a one-form equation in E , say $dD = \beta D + \sigma$ with $D = D_0 + D_1 r$, $\beta = \beta_0 + \beta_1 r$, $\sigma = \sigma_0 + \sigma_1 r$, separates into two coupled equations with rational coefficients,

$$dD_0 = \beta_0 D_0 + q \beta_1 D_1 + \sigma_0, \quad (94)$$

$$dD_1 + \frac{1}{2} D_1 d \log q = \beta_0 D_1 + \beta_1 D_0 + \sigma_1, \quad (95)$$

the even and odd components. Nothing algebraic has been lost and nothing irrational remains in the coefficients: the price of the root is a doubling of the number of unknowns. For k roots the same computation gives 2^k components, indexed by the subsets S , with the multiplication and differentiation rules displayed above.

7 The general solution: paths and polylogarithms

The canonical form reduces integration to the evaluation of the iterated integrals in (43) along paths in the physical region. For letters *linear* in the integration parameter the iterated integrals belong to a standard class:

Definition 7.1 (Generalized polylogarithms). For $a_1, \dots, a_k \in \mathbb{C}$ define recursively

$$G(a_1, a_2, \dots, a_k; t) = \int_0^t \frac{dt'}{t' - a_1} G(a_2, \dots, a_k; t'), \quad G(; t) = 1, \quad (96)$$

the integration running along a path from 0 to t that avoids the points a_i , with a branch of each logarithm fixed once along it; the value depends on the homotopy class of that path and on nothing else. The recursion is well posed unless every a_i vanishes, in which case the integral diverges at the lower limit; those words are defined instead by $G(\vec{0}_k; t) = \frac{1}{k!} \log^k t$, and the two prescriptions cover every word. These generalized polylogarithms [12] of weight k contain the classical functions: $G(0, 1; t) = -\text{Li}_2(t)$, and all functions of weight two and three reduce to classical polylogarithms.

Example 7.2 (From the definition to the dilogarithm). Two applications of the definition, for $0 < t < 1$:

$$G(1; t) = \int_0^t \frac{dt'}{t' - 1} = \log(1 - t), \quad G(0, 1; t) = \int_0^t \frac{dt'}{t'} G(1; t') = \int_0^t \frac{dt'}{t'} \log(1 - t') = -\text{Li}_2(t). \quad (97)$$

This is Example 4.2 read backwards: the two iterated integrals generated there by the canonical recursion are precisely $F_2^{(1)} = G(1; w)$ and $F_2^{(2)} = -G(0, 1; w)$.

Evaluating a family is then a matter of choosing a base point p_0 in the triangle (7) and a path to the point of interest, composed of segments on which every letter restricts to a polynomial of degree one in the path parameter. The criterion is exactly what the definition of G requires: if $\varphi_a(\gamma(t)) = \kappa_a(t - \alpha_a)$ then

$$d \log \varphi_a(\gamma(t)) = \frac{dt}{t - \alpha_a}, \quad (98)$$

the constant κ_a dropping out, so that every iterated integral of (43) along the segment is one of the functions G , with indices the roots α_a of the restricted letters. When a letter restricts to a quadratic, subdividing the segment does not help: the restriction stays quadratic on every piece. The remedy is either to choose the segment differently — in practice segments parallel to the coordinate axes, on which the bilinear letters produced by the rationalizing substitutions become linear again — or to factor the quadratic and admit its two linear factors, with algebraic constants, as letters in their own right. Flatness guarantees that the result depends only on the endpoint and on the homotopy class of the path; the zero locus of a letter is crossed only as a deliberate analytic continuation, never in passing. Families whose canonical form lives in rationalizing variables are evaluated in those variables throughout, along paths chosen by the same linearity criterion, and only the final values are mapped back to (v, w) .

Example 7.3 (Choosing the path). Suppose the alphabet of a family is $\{v, w, 1 - v - w, 1 - 4vw\}$ and the value at a point (v_1, w_1) of the triangle is wanted. Here $1 - 4vw$ occurs as a polynomial letter in its own right; that the same polynomial occurs elsewhere as a radicand, in (79), is a separate matter and plays no role in this example. Take the path in two segments parallel to the axes: first in v at fixed w , then in w at fixed $v = v_1$. On the second segment, parametrized by $w = t$,

$$v \rightarrow v_1, \quad w \rightarrow t, \quad 1 - v - w \rightarrow (1 - v_1) - t, \quad 1 - 4vw \rightarrow 1 - 4v_1 t : \quad (99)$$

the letter v is constant and drops out of the connection, and every remaining letter has degree one in t , so the iterated integrals along the segment are the functions $G(a_1, \dots, a_k; t)$ with indices

drawn from the letters' roots, $a_i \in \{0, 1 - v_1, 1/(4v_1)\}$, seeded by the values accumulated on the first segment. Had the diagonal segment $v = w = t$ been chosen instead, the last letter would restrict to $1 - 4t^2$ — quadratic, and still quadratic on every sub-segment — and one would either return to axis-parallel segments or admit its factors $1 \mp 2t$, in general with algebraic roots, as letters in their own right.

7.1 The order of expansion required

The cost of an iterated integral grows steeply with its weight, so one does not expand every master to the same depth; the required depth is dictated by where the master enters the cross section. The rule is arithmetic. The assembled combination is

$$\sigma = \sum_m c_m(\varepsilon) I_m(v, w; \varepsilon), \quad c_m(\varepsilon) = \sum_{j \geq -p_m} \varepsilon^j c_m^{(j)}, \quad (100)$$

with p_m the order of the pole that the coefficient of the m -th master carries after reduction and after the expansion of the measure. If σ is wanted through order ε^n , then I_m is wanted through order ε^{n+p_m} : the deepest pole a master multiplies anywhere in the combination sets its depth. Thus $\varepsilon^{-2}(w^\varepsilon - 1) = \varepsilon^{-1} \log w + \frac{1}{2} \log^2 w + \frac{1}{6} \varepsilon \log^3 w + \dots$ needs the ε^2 coefficient of w^ε for its finite part and no more. Cancellations of poles between terms do not relax the requirement, since each master must be known before the cancellation can be exhibited. In the double-real channel the outcome is lopsided: of the 347 masters, 203 are needed only to their finite order and 131 one order deeper, while 9 are needed deeper still and the remaining four less deeply (two enter only at order ε , and two multiply coefficients that vanish identically). This accounting is settled before any integration, and it sets the scale of the entire computation.

8 Boundary conditions

8.1 The space of solutions and the conditions upon it

The path-ordered solution (43) involves the undetermined vector $\vec{F}(p_0) = \sum_k \varepsilon^k \vec{C}^{(k)}$: one constant per master per order. Almost all of these are fixed by consistency alone, without evaluating a single new integral. The most productive condition — derived in the next subsection — is the *exclusion of local behaviour*: near each edge of the triangle the differential equation offers local modes with exponents read from its residue matrices (Section 4.3), the integral representation permits only some of them, and every offered mode that the representation forbids annihilates one combination of constants.

Three further conditions complete the analysis.

Elementary normalizations. The unit-power master of the lowest sector is the phase-space volume itself, a ratio of Γ -functions in closed form (Example 8.3); its Laurent expansion is the boundary datum of that sector wherever the sector occurs.

Inheritance. Families are analysed in an order compatible with the subsector relation. A master of a subsector, once evaluated, is literally the same integral in every family in which that subsector reappears — the identification of Section 3 is an equality of integrals, not an analogy. Expanding the known function at the base point of the new family and comparing with $\vec{I} = T\vec{F}$ there fixes the corresponding components of $T(p_0) \vec{F}(p_0)$, one inhomogeneous linear condition on the $\vec{C}^{(k)}$ per order.

Regularity at spurious loci. A letter may vanish at a locus interior to the domain in which a given master's integral representation converges, so that this master is finite there while the

equation still offers modes u^λ in the distance u to the locus. Every offered mode that is singular there — those with $\operatorname{Re} \lambda < 0$ for small ε , together with the powers of $\log u$ accompanying $\lambda = 0$ — must then have vanishing coefficient.

8.2 Power counting at the boundary

Consider a master given by a convergent cut phase-space integral, and examine one edge of the triangle, say $w \rightarrow 0$. In the limit the integral is a sum over momentum regions — hard, soft, collinear to one or the other direction — each of which is a homogeneous scaling of the integration momenta and therefore contributes with a definite overall power of w . That power is counted as follows. The regulator enters only through the measure, so the propagators of a region, each scaling as an integer power of w , contribute an integer, while a momentum taken soft, $k \sim w$, contributes

$$d^d k \sim w^d = w^{4-2\varepsilon}, \quad (101)$$

and a momentum taken collinear, $k^+ \sim 1$, $k^- \sim w$, $k_\perp \sim \sqrt{w}$, contributes

$$dk^+ dk^- d^{d-2} k_\perp \sim w^{d/2} = w^{2-\varepsilon}. \quad (102)$$

The exact function therefore has an expansion in the modes

$$w^{n+a\varepsilon}, \quad n \in \mathbb{Z}_{\geq n_0}, \quad (103)$$

in which

$$a = -2j_{\text{soft}} - j_{\text{coll}} \quad (104)$$

counts the momenta the region scales. The list of admissible a is therefore short and can be written down without any computation. The integrals of the double-real channel have two loop momenta, so $j_{\text{soft}} + j_{\text{coll}} \leq 2$, and the possible pairs $(j_{\text{soft}}, j_{\text{coll}})$ — $(0, 0)$, $(0, 1)$, $(1, 0)$ and $(0, 2)$, $(1, 1)$, $(2, 0)$ — give

$$a \in \{0, -1, -2, -3, -4\}, \quad (105)$$

six pairs but five values, since $(1, 0)$ and $(0, 2)$ both give $a = -2$; and five is exactly the spectrum of ε -coefficients observed on the soft edge over the whole channel. Which of the five actually occur for a given master is decided by its propagator content, since that is what decides which regions exist; the datum is coarse and common to whole groups of sectors. The differential equation, on the other hand, offers local modes with exponents drawn from the eigenvalues of the residue matrix at the corresponding letter: in the notation of Section 4.3, the general solution near $w = 0$ with residue matrix R_0 organizes itself as

$$\vec{F}(v, w) = \sum_{\lambda} w^{\lambda} \vec{f}_{\lambda}(v, \log w), \quad \lambda \in \varepsilon \operatorname{spec} R_0 + \mathbb{Z}_{\geq 0}, \quad (106)$$

with each $\vec{f}_{\lambda}$ polynomial in $\log w$ and linear in the boundary constants.

The two lists must be brought into one basis before they are compared, and they are not yet in one: the region analysis constrains the master $\vec{I} = T\vec{F}$, whereas (106) lists the exponents of the canonical $\vec{F}$, whose integer parts are non-negative by construction. If T has order ν at the edge — a pole of order $-\nu$ when $\nu < 0$ — then the exponents offered for the master are those of (106) shifted componentwise by ν ,

$$\lambda_{\vec{I}} \in \nu + \varepsilon \operatorname{spec} R_0 + \mathbb{Z}_{\geq 0}, \quad (107)$$

and it is this shifted list that the power counting constrains; it is also where the negative integer parts used in Section 9 come from. Whenever (107) offers an exponent that the power counting excludes, the entire corresponding mode must vanish, $\vec{f}_\lambda \equiv 0$: one exact, homogeneous, linear condition on the boundary constants per excluded mode. The same analysis applies at $v \rightarrow 0$ and on the soft edge $1 - v - w \rightarrow 0$, with the distance to the edge as the scaling variable.

Two cautions are essential in practice. The admissible exponents must be derived in the same variables in which the local analysis is performed, a rationalizing substitution altering the measure by its Jacobian and shifting the integer parts. And degenerate (Jordan) directions of the residue matrix couple modes logarithmically, so the exclusion argument must be applied to generalized eigenspaces, not to eigenvalues alone.

8.3 The kernel of the constraint system

All conditions of the preceding subsections are linear in the boundary constants of a given block and assemble, order by order in ε , into one system

$$C \vec{C}^{(k)} = \vec{d}^{(k)}, \quad (108)$$

homogeneous ($\vec{d} = 0$) for the excluded modes and the regularity conditions, inhomogeneous for the inherited values and the elementary normalizations. The structure is the affine-linear one already met at the completion equation: the solution set is an affine subspace, one particular solution plus $\ker C$, and the number of genuinely undetermined constants is $\dim \ker C$ — a property of the constraint matrix alone, known before a single integral is evaluated. Constants surviving in the kernel are then identified across blocks: the same subsector appearing in several families is the same integral, by the equality established in Section 3, so it contributes one constant and not several. For the double-real channel the constraint analysis carried out this way leaves at most thirty-three candidate periods for the whole problem, of which the lowest tier is established.

Definition 8.1 (Boundary period). *A boundary period is the value of a cut phase-space integral at a distinguished limit of the kinematics, the limits being taken in a definite order; equivalently, a constant of the general solution that no linear condition of the kind above determines.*

At the weights relevant here the boundary periods evaluate to $\mathbb{Q}$ -linear combinations of products of ζ -values and kindred constants. Each is established by an exact derivation — through a Mellin–Barnes representation, an expansion by momentum regions, or hypergeometric summation — with high-precision numerical evaluation used to corroborate, never to prove. The Mellin–Barnes representation rests on one identity, the splitting of a sum in the denominator,

$$\frac{1}{(X+Y)^\lambda} = \frac{1}{\Gamma(\lambda)} \frac{1}{2\pi i} \int_{-i\infty}^{+i\infty} dz \Gamma(\lambda+z) \Gamma(-z) \frac{X^z}{Y^{\lambda+z}}, \quad (109)$$

the contour separating the poles of $\Gamma(\lambda+z)$ from those of $\Gamma(-z)$. Applied to a propagator or to a sum of Feynman parameters it trades an integration over the integrand for a contour integral of Γ -functions, in which the parametric integrals can be done one at a time.

Example 8.2 (A period end to end). *The simplest ordered-limit integral of the whole subject is*

$$P(a, b) = \int_0^1 dx x^{-1+a\varepsilon} (1-x)^{-1+b\varepsilon} = \frac{\Gamma(a\varepsilon) \Gamma(b\varepsilon)}{\Gamma((a+b)\varepsilon)}, \quad (110)$$

Euler's beta integral, convergent for $\text{Re}(a\varepsilon)$ and $\text{Re}(b\varepsilon)$ positive and continued from there. Its expansion is obtained from $\log \Gamma(1+z) = -\gamma_E z + \sum_{k \geq 2} \frac{(-1)^k \zeta_k}{k} z^k$ after writing $\Gamma(z) = \Gamma(1+z)/z$:

$$P(a, b) = \frac{(a+b)\varepsilon}{ab\varepsilon^2} \frac{\Gamma(1+a\varepsilon)\Gamma(1+b\varepsilon)}{\Gamma(1+(a+b)\varepsilon)} = \frac{a+b}{ab\varepsilon} \left[1 - ab\zeta_2\varepsilon^2 + O(\varepsilon^3) \right], \quad (111)$$

since the terms linear in γ_E cancel between numerator and denominator and the ζ_2 coefficient is $\frac{1}{2}[a^2 + b^2 - (a+b)^2] = -ab$. A ζ -value has appeared at the first nontrivial order, from a Γ -function and nothing else; the periods of the calculation are of this character, one or two integrations deeper.

Example 8.3 (The phase-space volume). *The volume of the massless n -particle phase space in d dimensions is fixed by Lorentz invariance and dimensional analysis up to a constant, and iterating the two-body formula gives it in closed form:*

$$\begin{aligned} \int \prod_{i=1}^n d^d p_i \delta^{(+)}(p_i^2) \delta^d\left(q - \sum_i p_i\right) &= (q^2)^{(n-1)(d/2-1)-1} \left(\frac{\pi^{d/2-1}}{2}\right)^{n-1} \\ &\times \frac{\Gamma(d/2-1)^n}{\Gamma((n-1)(d/2-1)) \Gamma(n(d/2-1))}, \end{aligned} \quad (112)$$

a pure ratio of Γ -functions; other normalizations of the measure differ by powers of 2π only. (At $d=4$ and $n=3$ this reduces to the familiar $\Phi_3 = \pi^2 q^2/8$.) It is the unit-power master of the lowest sector, requires no differential equation, and supplies through its ε -expansion the exact ζ -values that inheritance propagates upward.

Example 8.4 (The two-dimensional system, concluded). *The system of Example 4.2 can be solved exactly in ε , which displays the whole accounting at once. The first component is constant, $F_1 = C_1$, and the second obeys*

$$\partial_w F_2 = -\frac{\varepsilon}{w} F_2 - \frac{\varepsilon}{1-w} C_1. \quad (113)$$

Its homogeneous solution is $w^{-\varepsilon}$, and the particular solution regular at the origin is found by inserting $F_2 = \sum_{n \geq 1} a_n w^n$ and $-\varepsilon C_1(1-w)^{-1} = -\varepsilon C_1 \sum_{m \geq 0} w^m$, which gives $n a_n = -\varepsilon a_n - \varepsilon C_1$, so that

$$F_2 = C_2 w^{-\varepsilon} - \varepsilon C_1 \sum_{n \geq 1} \frac{w^n}{n + \varepsilon}. \quad (114)$$

Expanding the sum in ε reproduces $\varepsilon C_1 \log(1-w) + \varepsilon^2 C_1 \text{Li}_2(w) + O(\varepsilon^3)$, in agreement with Example 4.2. The two boundary constants are visible in (114): C_2 multiplies the mode $w^{-\varepsilon}$ offered by the eigenvalue -1 of the residue matrix at $w=0$, and C_1 multiplies the mode w^0 offered by the eigenvalue 0 . The exclusion of Section 8.2 can now be carried out in full. Suppose the integral represented by F_2 has, in the limit $w \rightarrow 0$, only the hard region, so that (104) gives $a=0$ and its admissible exponents are the non-negative integers alone. The equation offers

$$\varepsilon \text{spec } R_0 + \mathbb{Z}_{\geq 0} = \{0, -\varepsilon\} + \mathbb{Z}_{\geq 0}, \quad (115)$$

so the mode $w^{-\varepsilon}$ is offered and forbidden, and its coefficient must vanish: $C_2 = 0$. Note that the argument is the comparison of exponent lists, not an appeal to finiteness — $w^{-\varepsilon}$ is itself bounded, and indeed vanishing, as $w \rightarrow 0$ for $\text{Re } \varepsilon < 0$, so finiteness alone would exclude nothing. The remaining constant C_1 is inherited, the first component being the volume-type subsector master already known. The kernel of the constraint system is trivial, no new period appears, and $\text{Li}_2(w)$ carries no free constant — in miniature, the accounting of the full calculation.

9 Behaviour at the edges of phase space

Section 4.3 identified the exponents and the mode structure at each singular locus, and Section 8 fixed the constants that multiply them. One further step converts the modes into the objects the cross section needs.

Fix an edge of the triangle and let u denote the distance to it: $u = 1 - v - w$ on the soft edge, where this matters most since the Born configuration sits there, and $u = v$ or $u = w$ on the collinear edges, where the same identities apply. Along the soft edge one variable survives, say v , with $w = 1 - v - u$. The master then has, near that edge, the expansion

$$\vec{I}(v, w) = \sum_{n,a} u^{n+a\varepsilon} \vec{g}_{n,a}(v) + \dots, \quad (116)$$

the dots standing for the accompanying powers of $\log u$, and the cross section integrates it against functions smooth up to the edge. The coefficients are obtained in two steps. Projecting the canonical solution onto the Frobenius basis (48) gives the coefficient of each canonical mode: with R_u the residue matrix at the edge,

$$\lim_{u \rightarrow 0} \Pi_\lambda u^{-\varepsilon R_u} \vec{F}, \quad (117)$$

Π_λ the projector onto the generalized eigenspace of exponent λ , the limit existing because $H(u) \rightarrow 1$, and for generic small ε — the non-resonance regime of Section 4.3, $|\operatorname{Re} \varepsilon(\lambda_i - \lambda_j)| < 1$, so that the modes do not mix under the limit. The $\vec{g}_{n,a}$ of (116) are then the Laurent coefficients of T at the edge contracted with these canonical mode coefficients, and it is that step which produces the exponents $n \leq -1$: the canonical $\vec{F}$ has non-negative integer parts by construction, and the negative ones come from the pole of T , exactly as Section 8.2 records. On the soft edge the $\vec{g}_{n,a}$ are functions of v , not numbers, and it is those functions which multiply the distributions below.

For such a mode the expansion in ε and the integration against a test function do not commute. The correct object is the distributional identity

$$u^{-1+a\varepsilon} = \frac{1}{a\varepsilon} \delta(u) + \sum_{k \geq 0} \frac{(a\varepsilon)^k}{k!} \left[\frac{\log^k u}{u} \right]_+, \quad (118)$$

where the *plus distribution* is defined by

$$\int_0^1 du [g(u)]_+ f(u) = \int_0^1 du g(u) [f(u) - f(0)] \quad (119)$$

for test functions f smooth at $u = 0$. The proof of (118) is one line: integrate against f , add and subtract $f(0)$,

$$\int_0^1 du u^{-1+a\varepsilon} f(u) = \frac{f(0)}{a\varepsilon} + \int_0^1 du u^{-1+a\varepsilon} [f(u) - f(0)], \quad (120)$$

where the first term uses $\int_0^1 u^{-1+a\varepsilon} du = 1/(a\varepsilon)$, convergent for $\operatorname{Re}(a\varepsilon) > 0$ and continued from there; the second integrand may be expanded in ε term by term because $f(u) - f(0)$ vanishes at $u = 0$, making every $u^{-1} \log^k u [f(u) - f(0)]$ integrable. Modes $u^{n+a\varepsilon}$ with $n \leq -2$ are treated by the same subtraction carried to higher order and produce derivatives of $\delta(u)$: for $n = -2$, subtracting the first *two* Taylor terms of the test function gives

$$u^{-2+a\varepsilon} = \frac{\delta(u)}{a\varepsilon - 1} - \frac{\delta'(u)}{a\varepsilon} + \sum_{k \geq 0} \frac{(a\varepsilon)^k}{k!} \left[\frac{\log^k u}{u^2} \right]_{++}, \quad (121)$$

with the double subtraction defined, in parallel with the single one, by

$$\int_0^1 du [g(u)]_{++} f(u) = \int_0^1 du g(u) [f(u) - f(0) - u f'(0)], \quad (122)$$

and $\delta'(u)$ acting as $\int_0^1 du \delta'(u) f(u) = -f'(0)$. The derivation is the same one-line computation, now using $\int_0^1 u^{-2+a\varepsilon} du = 1/(a\varepsilon - 1)$, convergent for $\text{Re}(a\varepsilon) > 1$ and continued from there.

As printed, (118) is exact on the collinear edges, where u is v or w and the range is $(0, 1)$. On the soft edge the range of $u = 1 - v - w$ at fixed v is $(0, 1 - v)$, and the endpoint enters:

$$\int_0^L du u^{-1+a\varepsilon} = \frac{L^{a\varepsilon}}{a\varepsilon}, \quad L = 1 - v, \quad (123)$$

so that the coefficient of $\delta(u)$ carries a factor $(1 - v)^{a\varepsilon}$, itself to be expanded in the regulator afterwards; the plus distributions acquire the corresponding rescaling. The identity is therefore applied at fixed v , with this factor absorbed into the coefficient functions $\vec{g}_{n,a}(v)$ of (116).

Had one expanded the *mode* first, the left-hand side would have become the series

$$\frac{1}{u} \sum_{k \geq 0} \frac{(a\varepsilon)^k}{k!} \log^k u, \quad (124)$$

each term of which is non-integrable at $u = 0$: the δ -distribution — the very content the cross section needs — is destroyed by premature expansion. For this reason the analytic evaluation carries the exact local data (116) next to the expanded polylogarithmic form of each master. The identity (118) is applied to the unexpanded modes, and only afterwards is everything expanded in the regulator. No resummation is involved anywhere in this step: the residue matrices supply the exponents, the boundary analysis supplies the coefficients, and (118) converts the two into distributions.

10 Search and proof

Nothing in the construction depends on how a transformation was found. An ansatz, a numerical fit, arithmetic modulo a prime — each produces a candidate and nothing more; what makes the candidate a theorem is the identity

$$T^{-1} A T - T^{-1} dT = \varepsilon \sum_a R_a d \log \varphi_a, \quad (125)$$

verified exactly in both variables. The same asymmetry runs through the whole subject: flatness establishes a connection, and a boundary period enters only through a derivation in which numerics appear afterwards, as a check on arithmetic, never as evidence. Where a verification is performed modulo primes at random points, its failure probability is bounded and stated, and the bound is elementary: a nonzero polynomial of total degree D vanishes at a point drawn uniformly from $\mathbb{F}_p^n$ with probability at most D/p (the Schwartz–Zippel lemma), so a false identity of degree at most D survives r independent evaluations with probability at most

$$\left(\frac{D}{p}\right)^r. \quad (126)$$

For a sixty-three-bit prime, $p > 4 \cdot 10^{18}$, and identities of degree $D \leq 10^4$, this is $(D/p)^2 < 10^{-29}$ already at $r = 2$. It is the stated bound — not an impression of reliability — that the result carries.

It is honest to record the state of the calculation these notes describe. At the time of writing, all but the three three-root families carry canonical forms established in this way — eighty-eight of the ninety-one — while the three-root families await the Galois-algebraic treatment of Section 6.4. Of the boundary periods, the lowest tier is established and the higher ones are not; nor is the local mode analysis of every master at every edge. The method is settled: each step above is defined and each of its claims is decided by an exact identity. What remains is its execution on the families and the periods still open.

11 Notation and terminology

Integral family / representative family. A complete set of inverse propagators (11) / its representative modulo cut-preserving redefinitions of the loop momenta.

Sector, subsector. The subset of propagators present with positive power; a subsector arises by deleting lines.

Master integral. Basis element of a family's integrals modulo integration-by-parts identities.

Connection, flatness. The matrix one-form A in $d\vec{I} = A\vec{I}$; the integrability condition $dA = A \wedge A$.

Block, class. A strongly connected component of the coupling graph of a system (a diagonal block of the triangular connection); its equivalence class under basis permutation composed optionally with $v \leftrightarrow w$.

Canonical form (ε -form), letters, alphabet, residues. The form $A = \varepsilon \sum_a R_a d \log \varphi_a$ with constant R_a ; the functions φ_a ; their set; the matrices R_a .

Completion equation. The linear equation (69) for the entries below the diagonal.

Root content, rationalizing substitution. The rank of the group of square classes in an alphabet; a rational change of variables under which roots become rational.

Galois algebra of the roots. The extension $\mathbb{Q}(v, w, \varepsilon)[r_1, \dots, r_k]/(r_i^2 - q_i)$ with automorphism group $(\mathbb{Z}/2)^k$ acting by sign changes.

Boundary period. A constant surviving all boundary conditions; the value of a cut phase-space integral at an ordered limit.

Local modes. The exponents $n + a\varepsilon$ and logarithm powers of the Frobenius solutions at a singular locus, read from the residue matrix.

Further reading. Of the references below, Henn's paper [3] introducing the canonical form is short and repays being read first, and Duhr's lectures [5] are the standard entry to iterated integrals and polylogarithms, including the shuffle and stuffle structure these notes pass over. The others are the sources of the individual constructions used above. Lee's Libra [7] and Meyer's CANONICA [8] are the standard implementations of the reduction to canonical form.

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